

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	Shaik Ashraf
Register Number	192324088

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I Shaik Ashraf/192324088 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:  _____

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Sol:

Aim: To create a simple cloud software application for Library Book Reservation System for SIMATS Library using Zoho Creator to demonstrate Software as a Service (SaaS).

Procedure:

Step 1: Go to www.zoho.com.

Step 2: Log in to your Zoho account.

Step 3: Open Zoho Creator and select Create New Application.

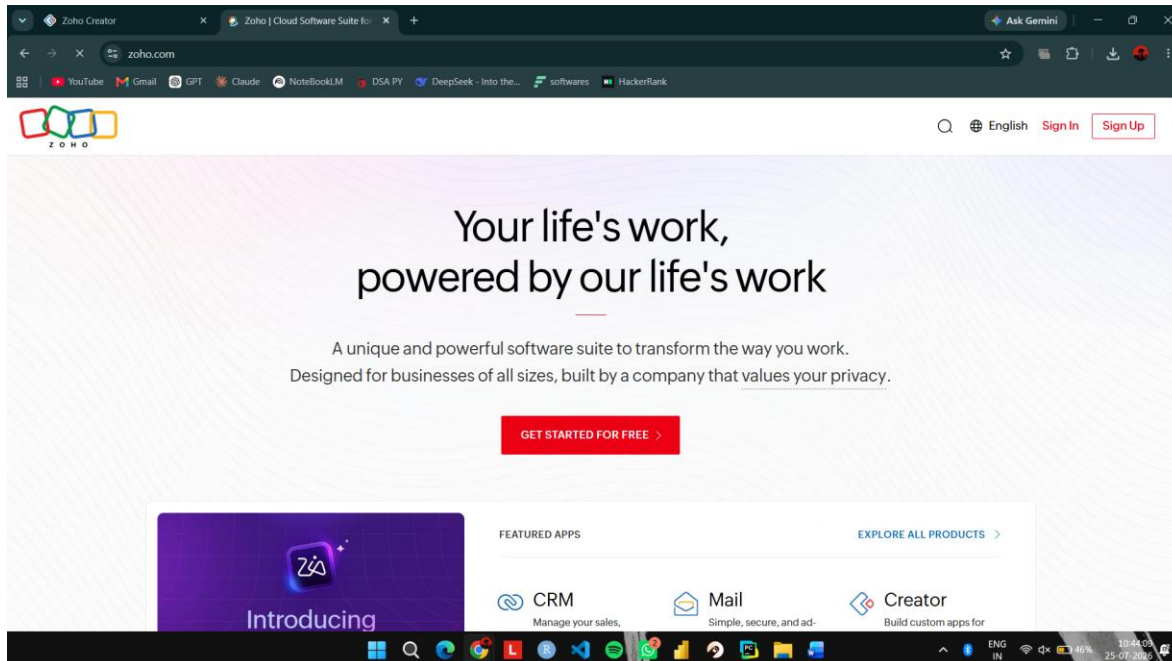
Step 4: Enter the application name as **Library Book Reservation System**.

Step 5: Create the required forms, reports, and a basic dashboard for library book reservation.

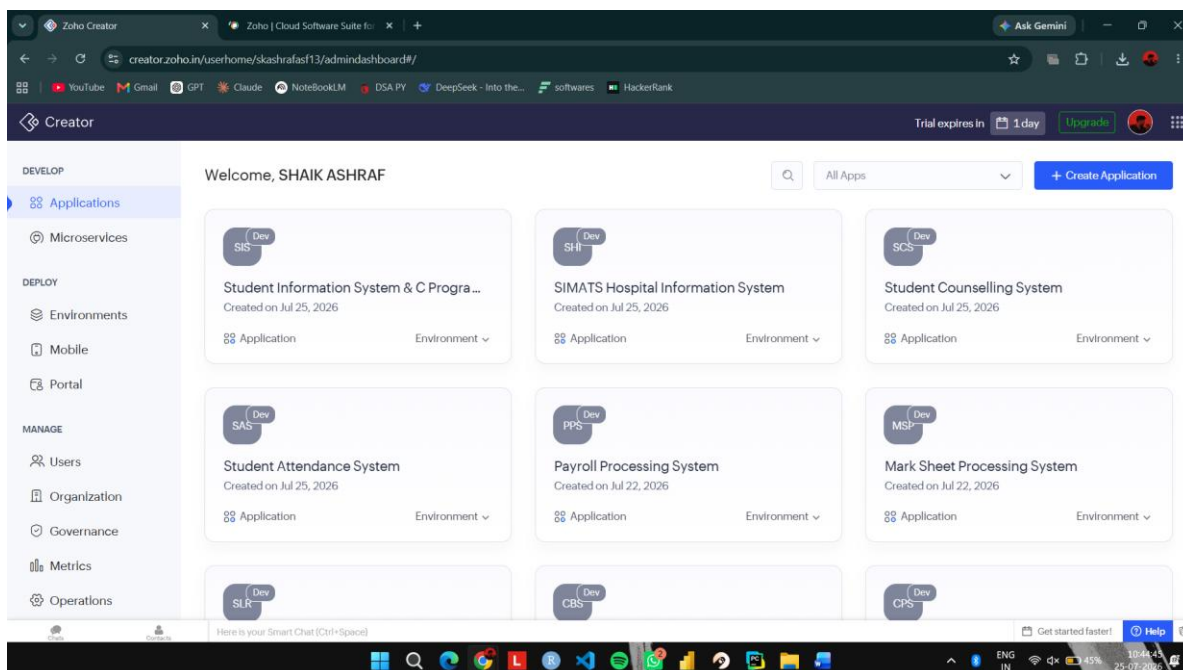
Step 6: Save and publish the application. The software has been created successfully.

IMPLEMENTATION:

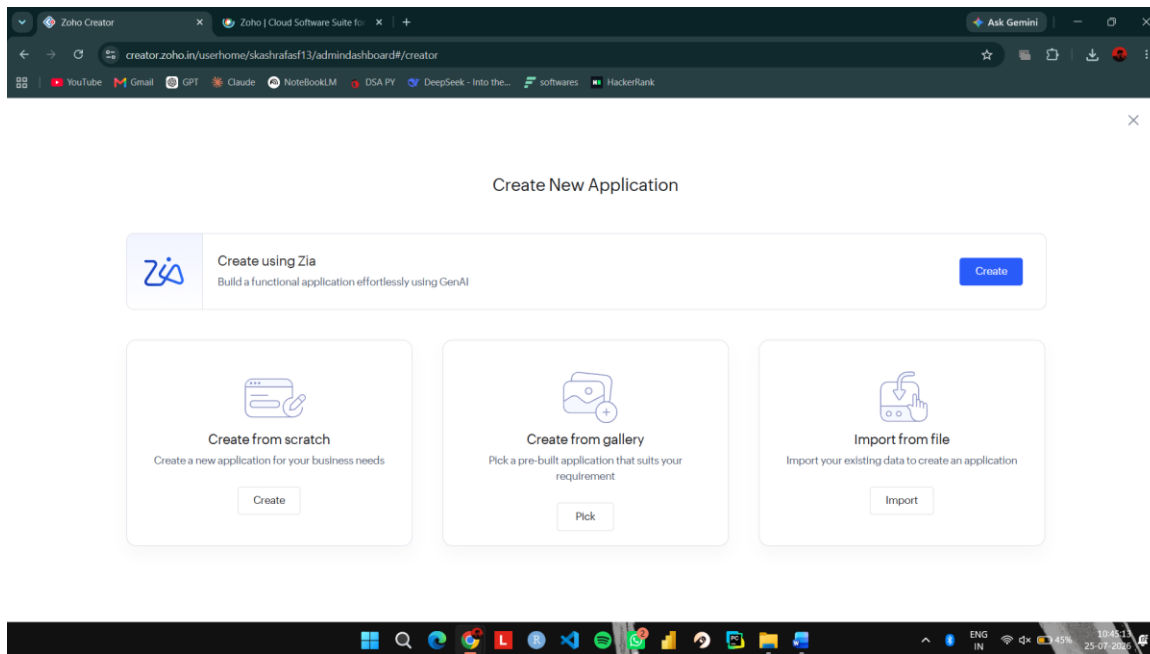
Step 1: Go to www.zoho.com



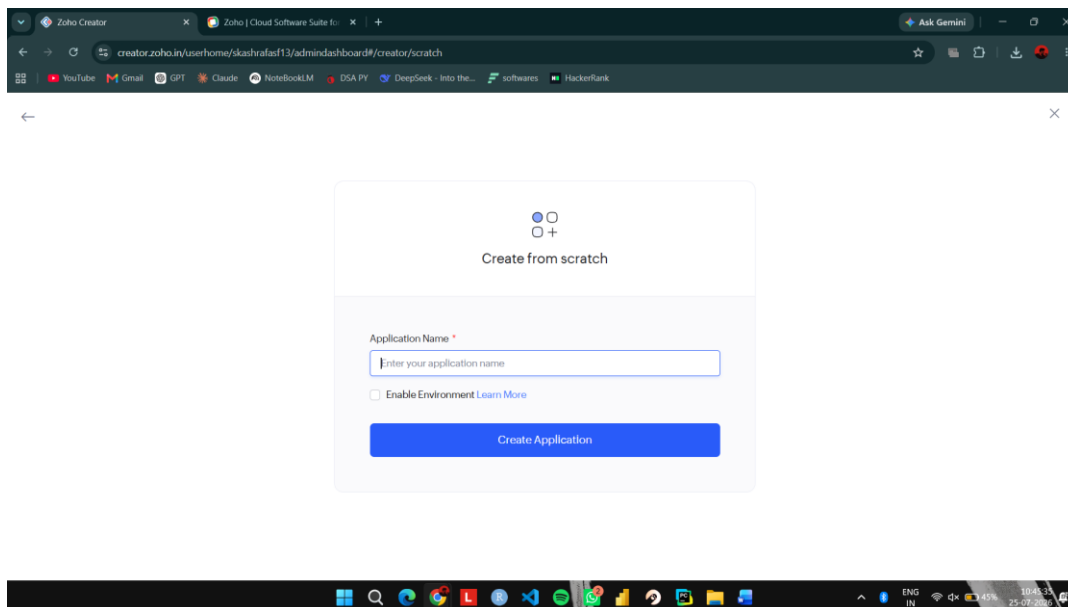
Step 2: Log in to your Zoho account.



Step 3: Open Zoho Creator and select Create New Application.



Step 4: Enter the application name as Mark Sheet Processing System.



Step 5: Create the required forms, reports, and a basic dashboard for library book reservation.

The screenshot shows the Zoho Creator form builder interface for a 'Library Book Reservations' application. On the left, there's a 'Basic Fields' panel with icons for Name, Email, Address, Phone, Single Line, Multi Line, Number, Date, Time, and Drop Down. The main area displays a form with fields: Book (with a search icon), Student Name, Student Register Number, Issue Date, Return Date, Status, and Created Date. On the right, the 'Field Properties' panel is open for the 'Book' field, showing its name, link name ('book'), display fields (Library Book Reservation → Books), display type (Multi Select), and validation options (Mandatory, Choices).

Step 6: Save and publish the application. The software has been created successfully.

The screenshot shows the published application dashboard. It features a navigation bar with 'Dashboard', 'Book Reservations', and 'Books'. The dashboard includes four summary cards: 'Count of Returned' (1), 'Count of Issued' (2), 'Count of Reserved' (2), and 'Count of Book Reservations' (5). Below these is a 'Status' section showing '2 Returned', '1 Issued', '2 Available', and '5 Total'. The main section is a 'Book Reservations Report' table.

Book	Student Name	Student Register Number	Issue Date	Return Date	Status	Created Date
2	Eva Chen	STU10005	19-Jul-2026	26-Jul-2026	Reserved	19-Jul-2026
3						
4						
1	David Lee	STU10004	18-Jul-2026	25-Jul-2026	Reserved	18-Jul-2026
2						
3						
4						
5						
1	Carol Morrison	STU10003	17-Jul-2026	24-Jul-2026	Issued	17-Jul-2026

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org, over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Sol:

Aim:

To create a simple cloud software application for Payroll Processing System using Zoho Creator to demonstrate Software as a Service (SaaS).

Procedure:

Step 1: Go to www.zoho.com.

Step 2: Log in to your Zoho account.

Step 3: Open Zoho Creator and select Create New Application.

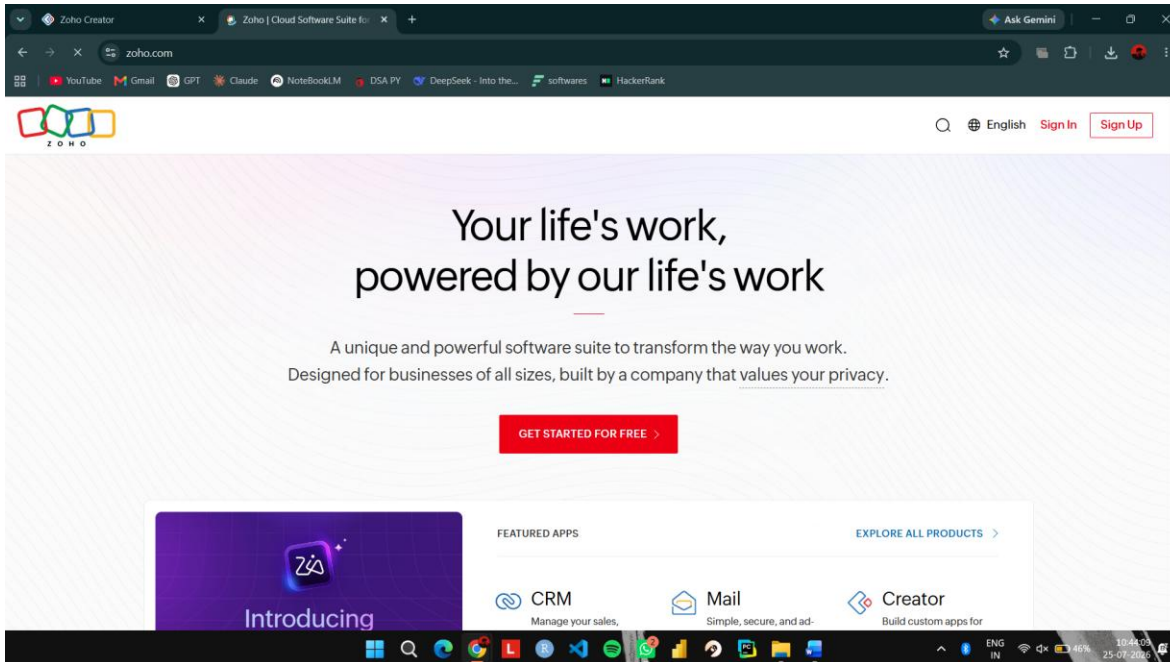
Step 4: Enter the application name as Payroll Processing System.

Step 5: Create the required forms, reports, and dashboard.

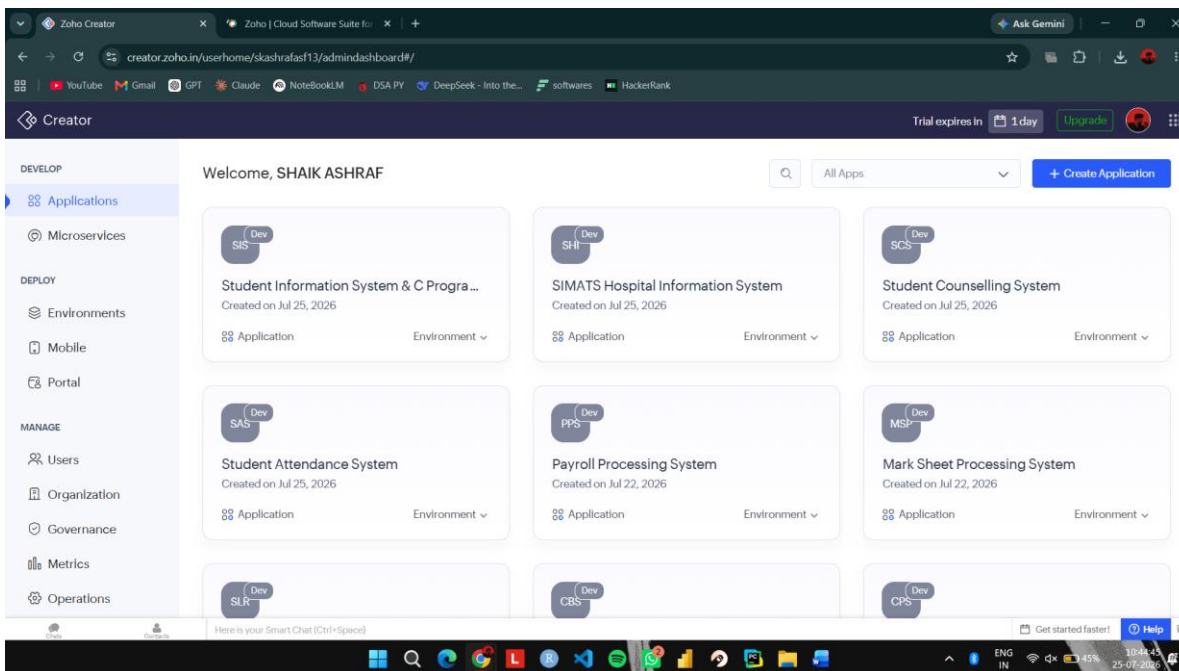
Step 6: Save and publish the application. The software has been created successfully.

IMPLEMENTATION:

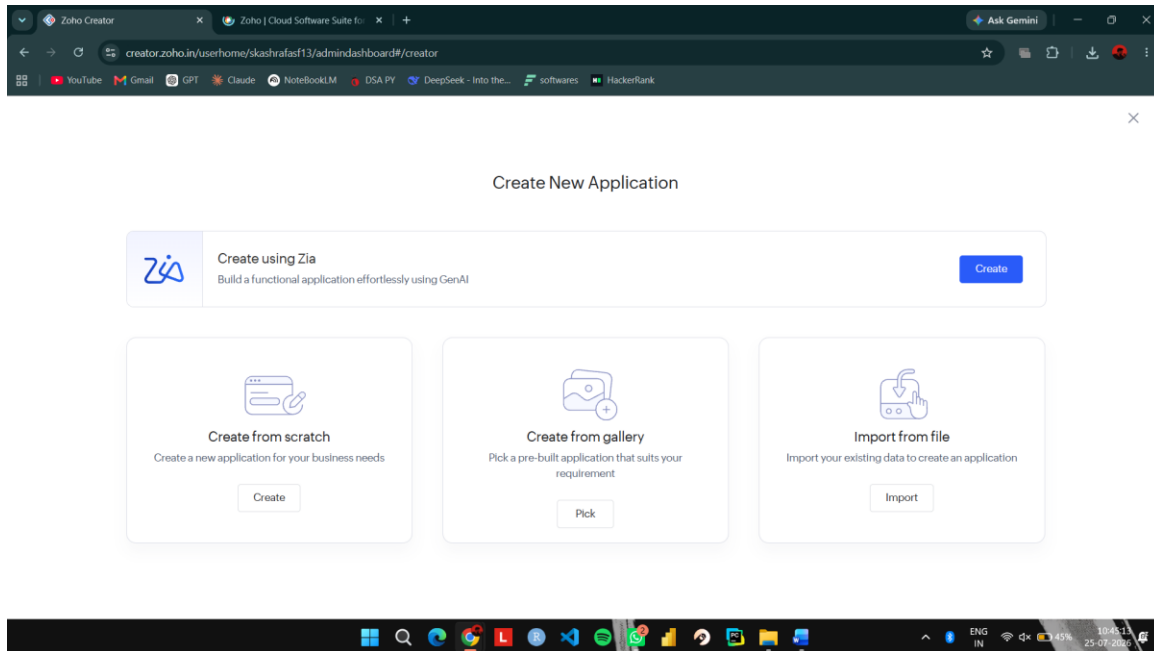
Step 1: Go to www.zoho.com.



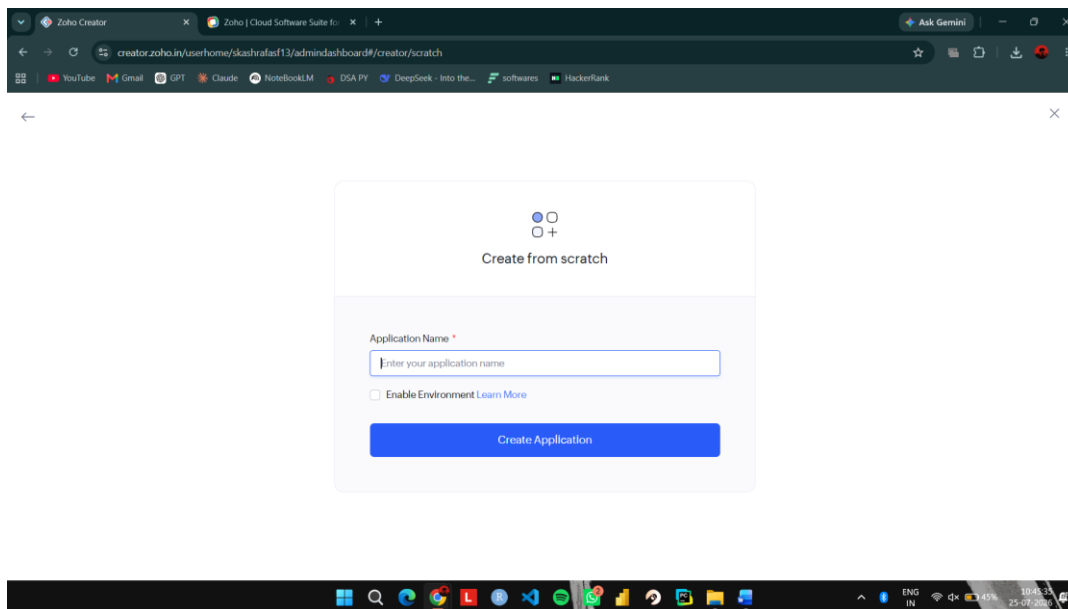
Step 2: Log in to your Zoho account.



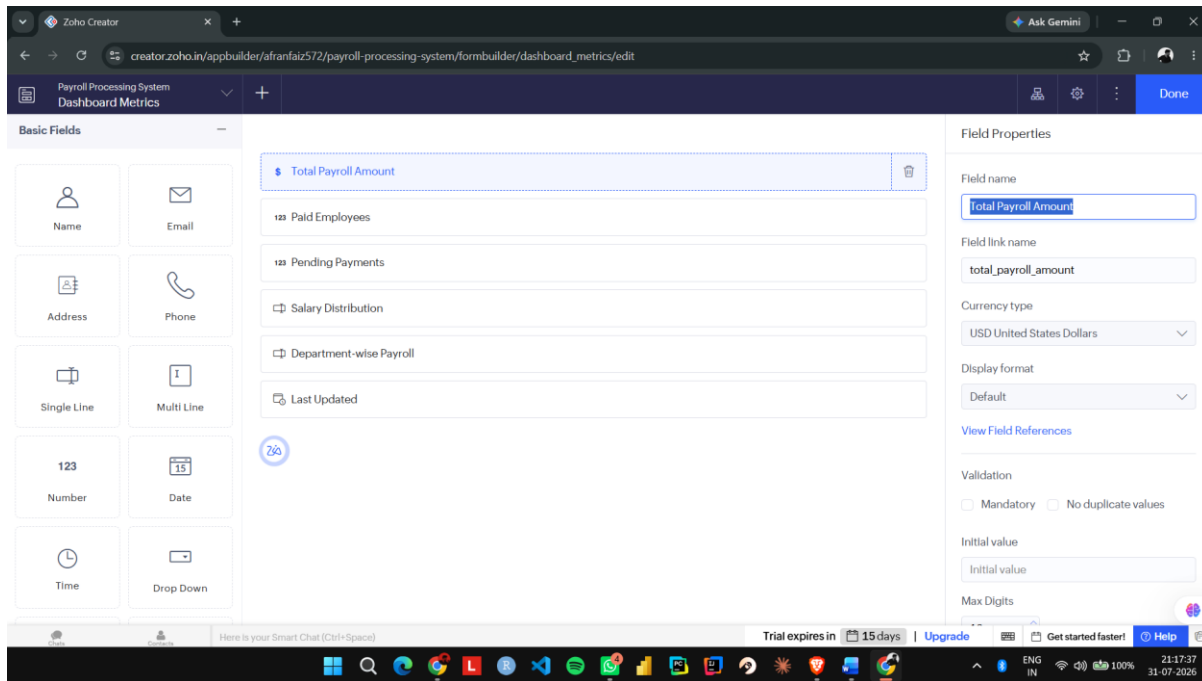
Step 3: Open Zoho Creator and select Create New Application.



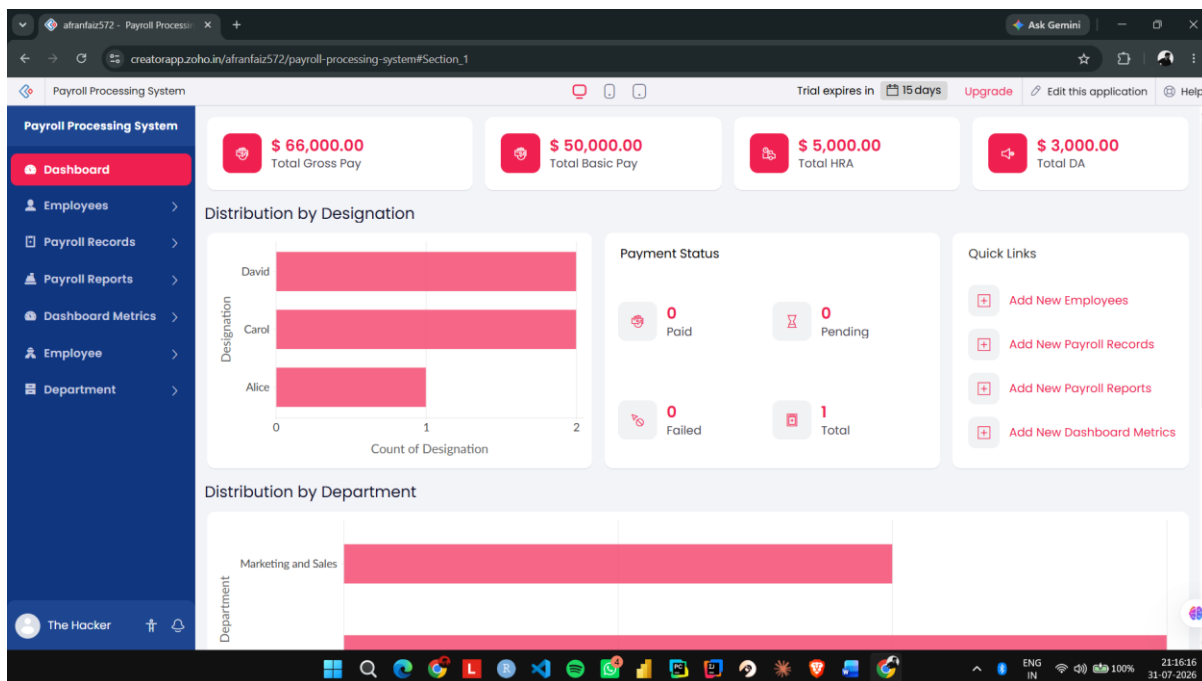
Step 4: Enter the application name as Payroll Processing System.



Step 5: Create the required forms, reports, and dashboard.



Step 6: Save and publish the application. The software has been created successfully.



Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Sol:

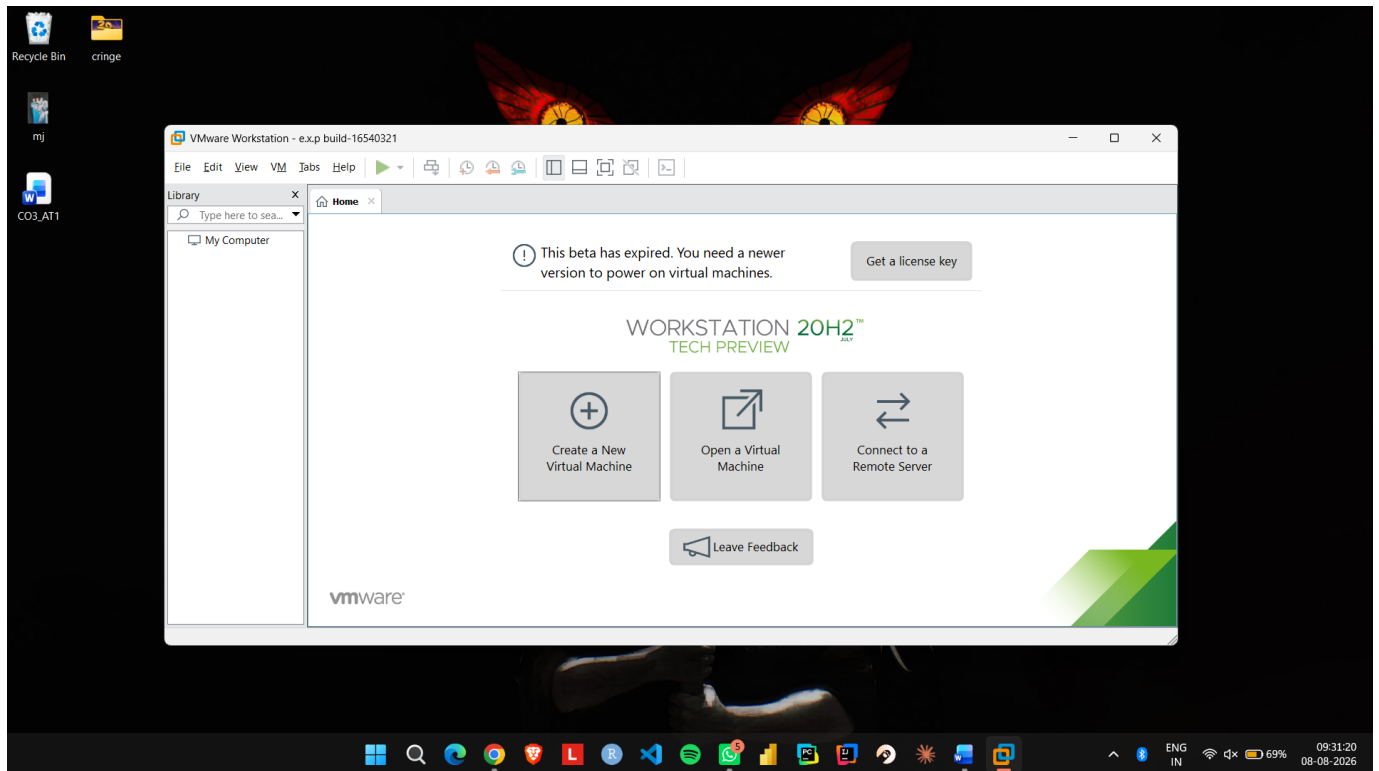
Aim: To create a virtual machine using **VMware Type-2 Hypervisor** with 1 CPU, 2 GB RAM and 15 GB storage.

Procedure

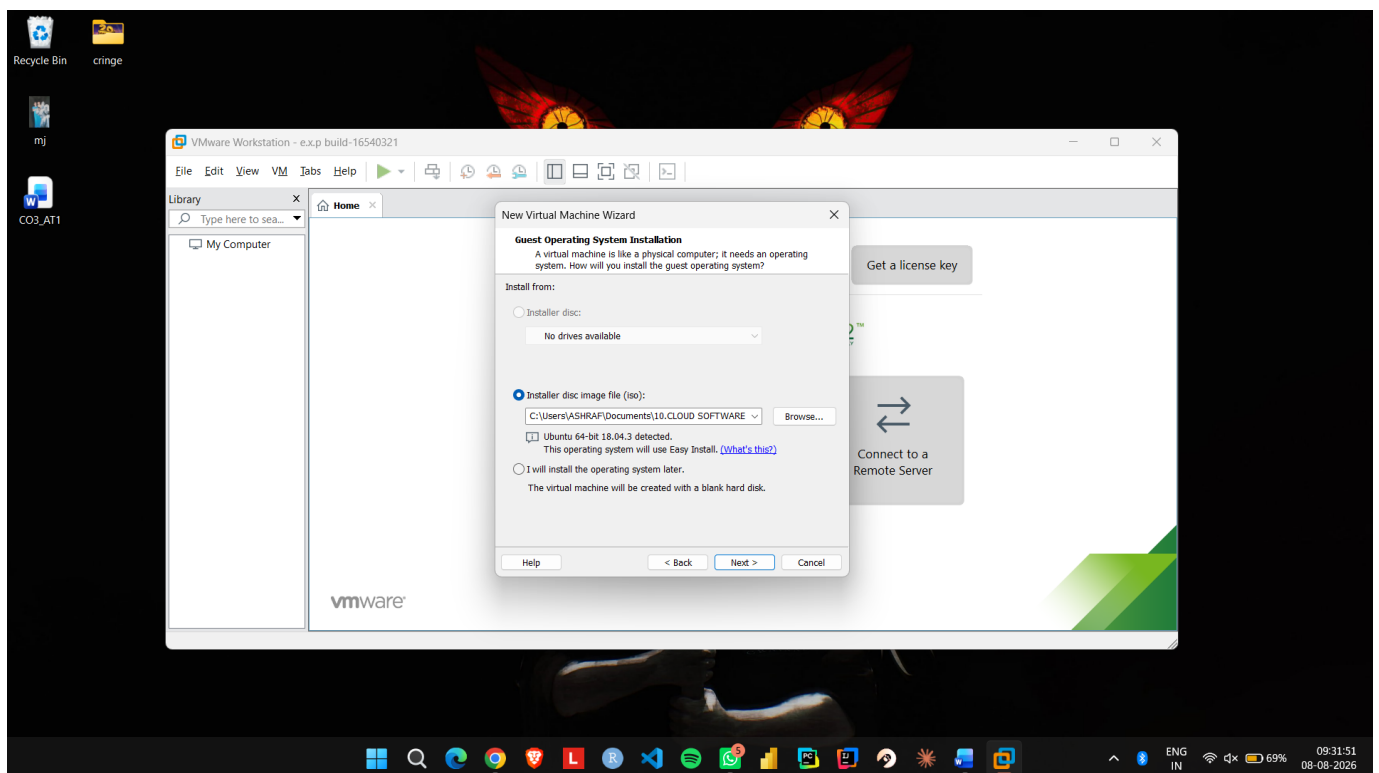
1. Open VMware and select Create New Virtual Machine.
2. Select the Ubuntu ISO file.
3. Set the VM name as Ubuntu-VM.
4. Set RAM = 2 GB and CPU = 1.
5. Set the hard disk size = 15 GB.
6. Start the VM and install Ubuntu.

IMPLEMENTATION:

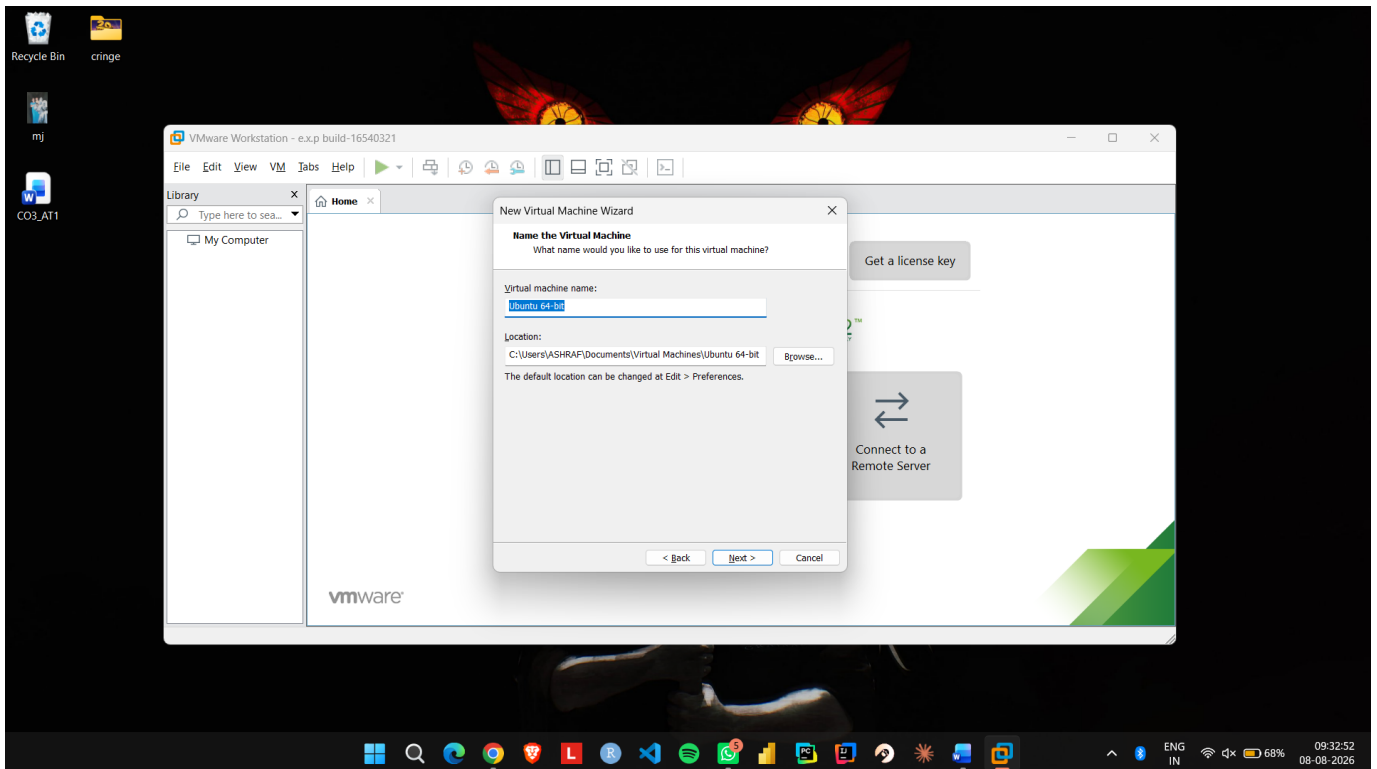
STEP 1: Open VMware and select Create New Virtual Machine.



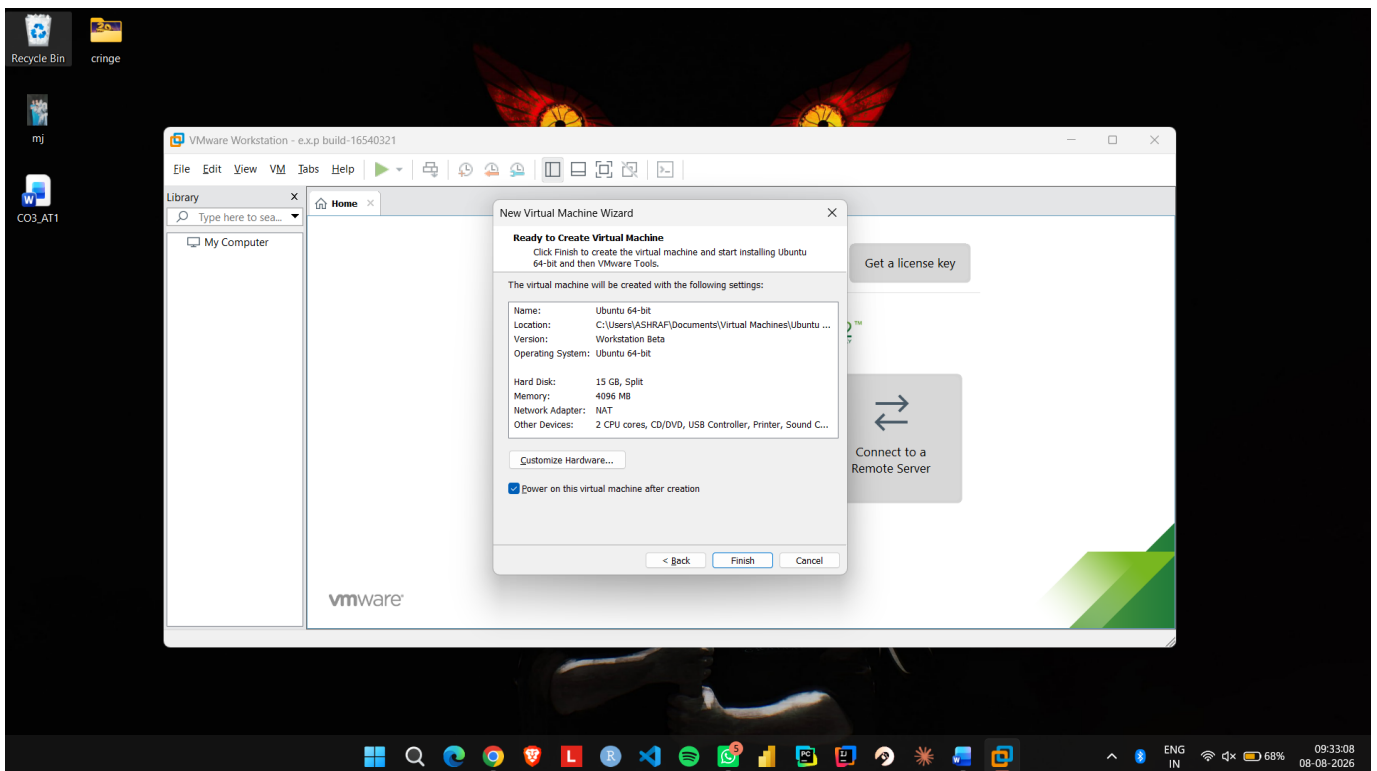
STEP 2: Select the Ubuntu ISO file.



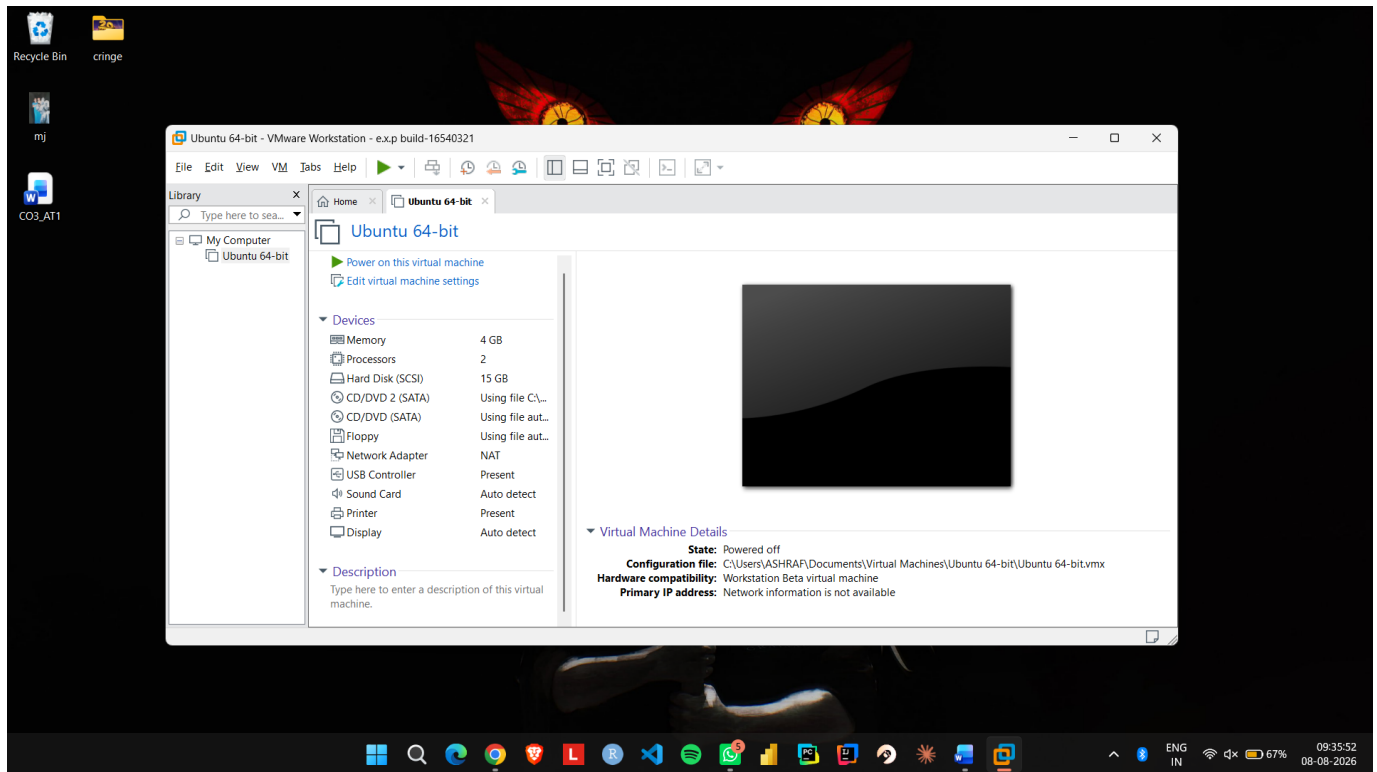
STEP 3: Set the VM name as Ubuntu-VM.



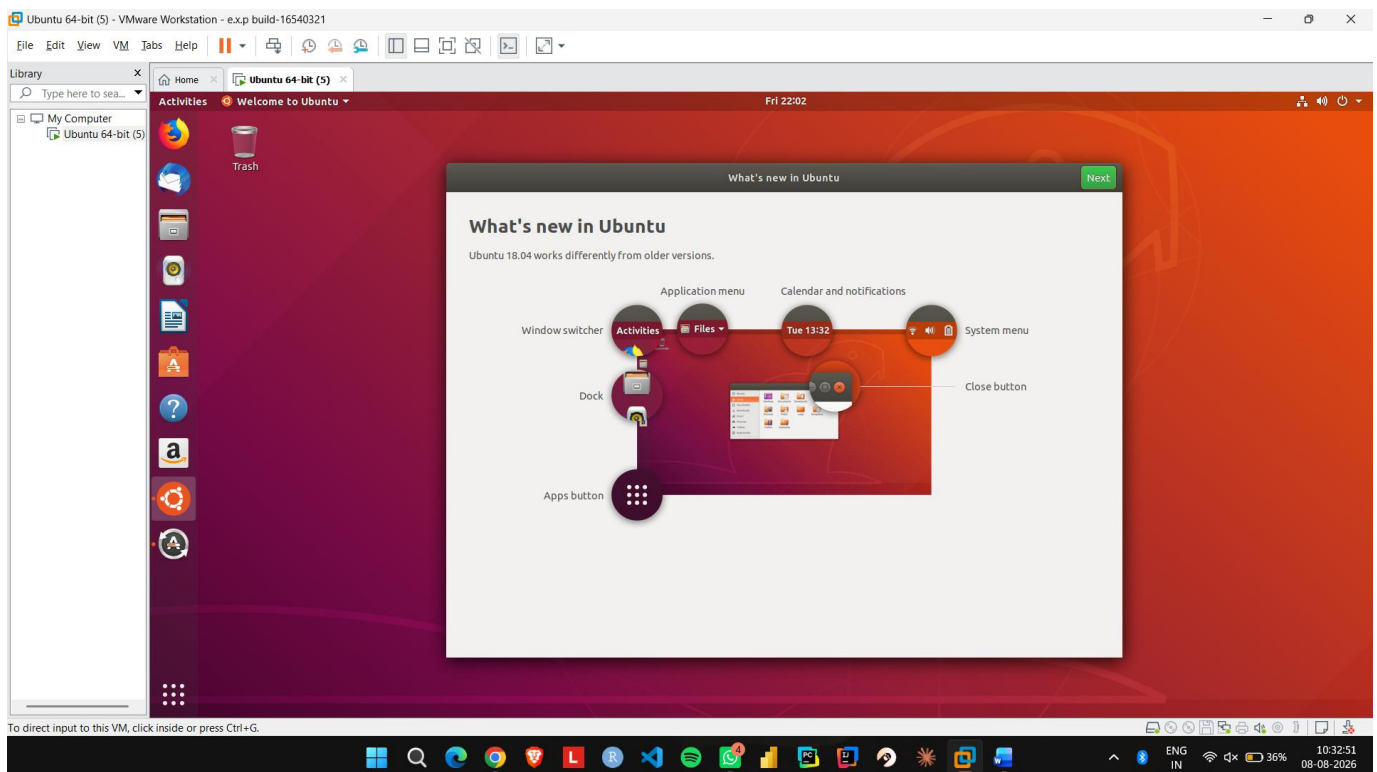
STEP 4: Set RAM = 2 GB and CPU = 1.



STEP 5: Set the hard disk size = 15 GB.



STEP 6: Start the VM and install Ubuntu



Result

The virtual machine was successfully created using VMware with 1 CPU, 2 GB RAM and 15 GB storage.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Sol:

Aim

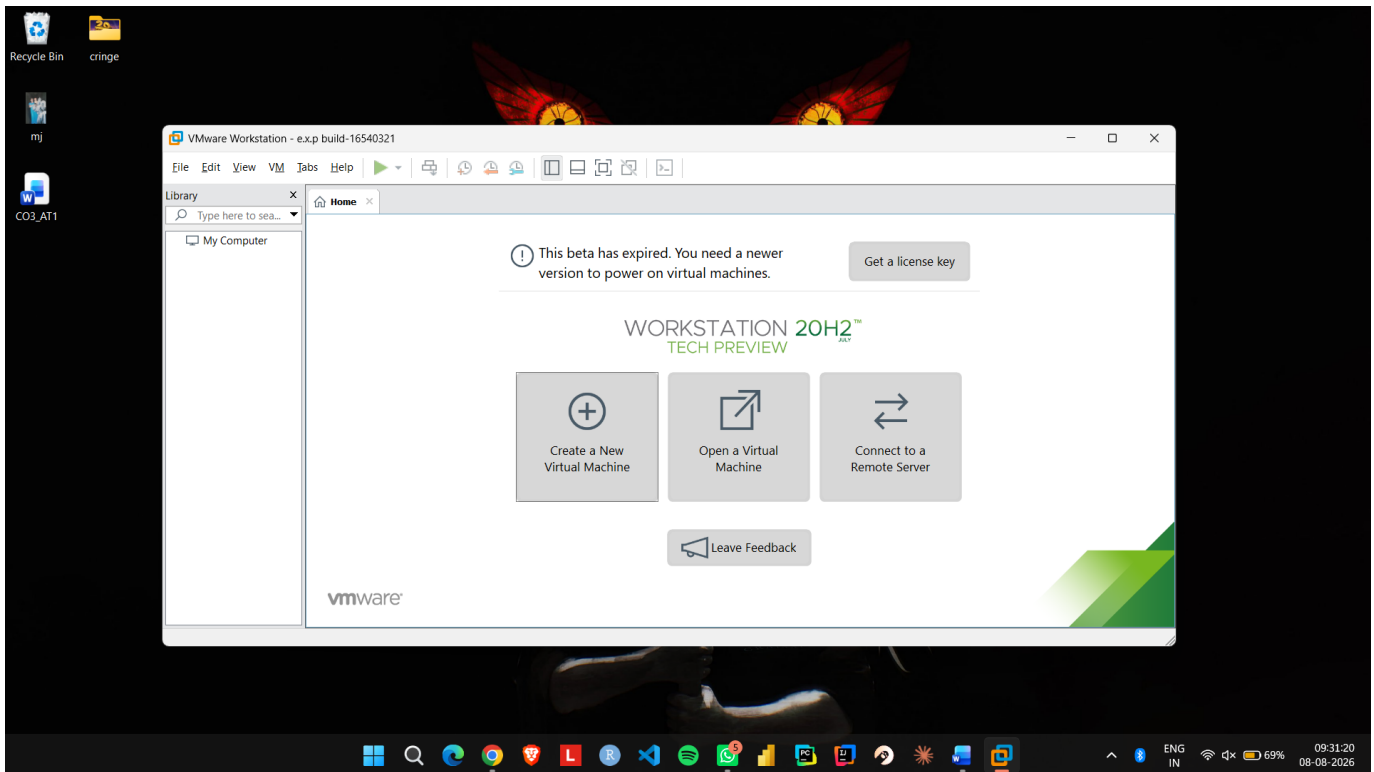
To create a clone of an existing virtual machine and test the cloned VM by restoring its previous version.

Procedure

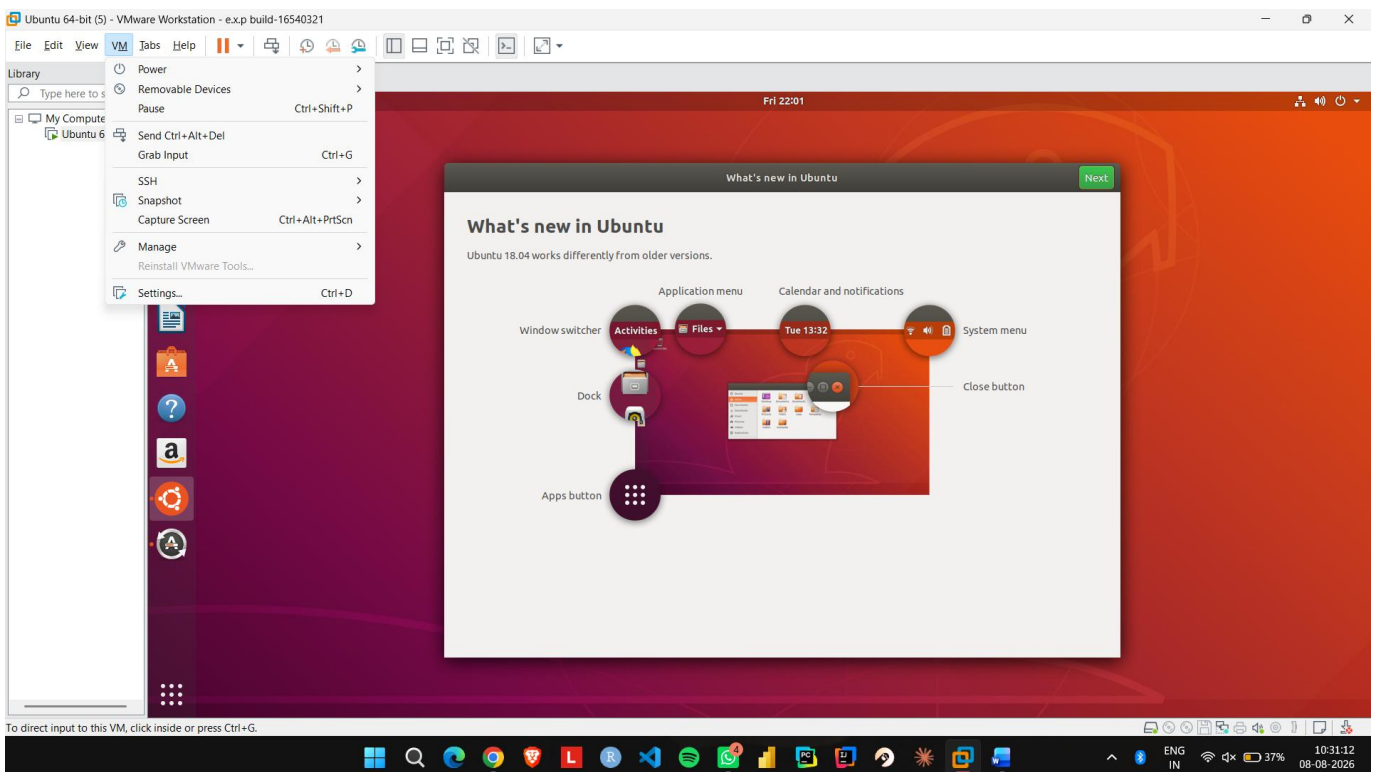
1. Open VMware Workstation, select the Ubuntu VM, and shut it down.
2. Select Manage → Clone to start the cloning process.
3. Select Full Clone and enter a name for the cloned VM.
4. Complete the cloning process and start the cloned VM.
5. Create a snapshot, make some changes, and then shut down the VM.
6. Restore the previous snapshot, start the VM, and verify that the previous state is loaded.

IMPLEMENTATION:

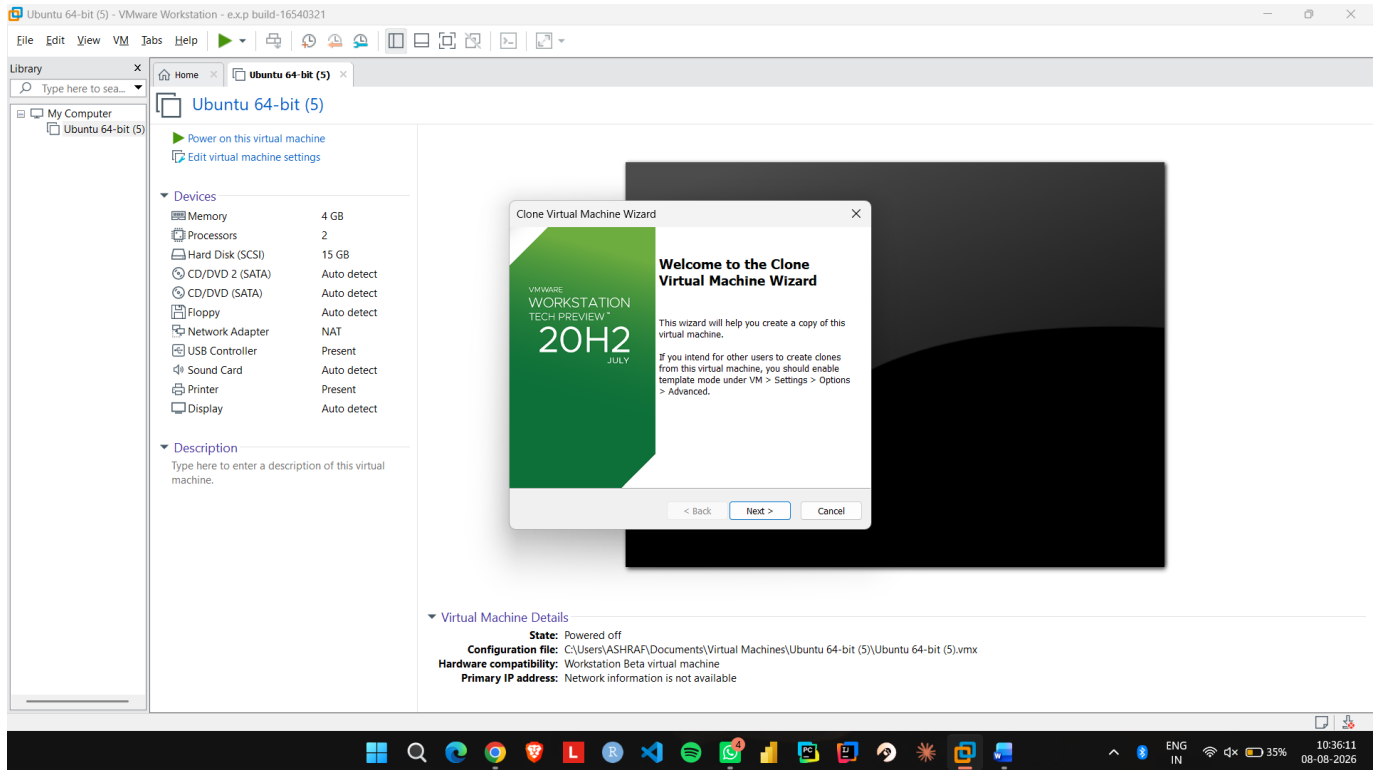
STEP 1: Open VMware Workstation, select the Ubuntu VM, and shut it down.



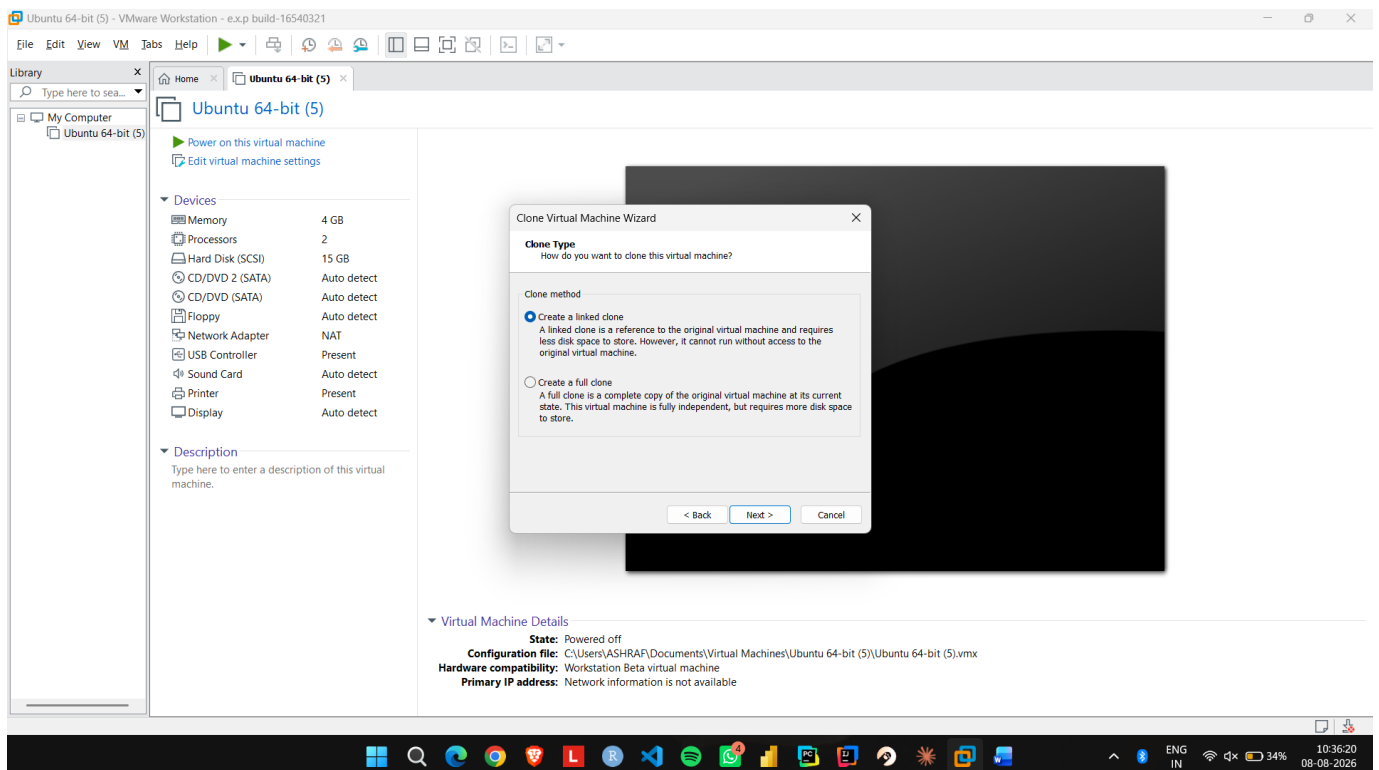
STEP 2: Select Manage → Clone to start the cloning process.



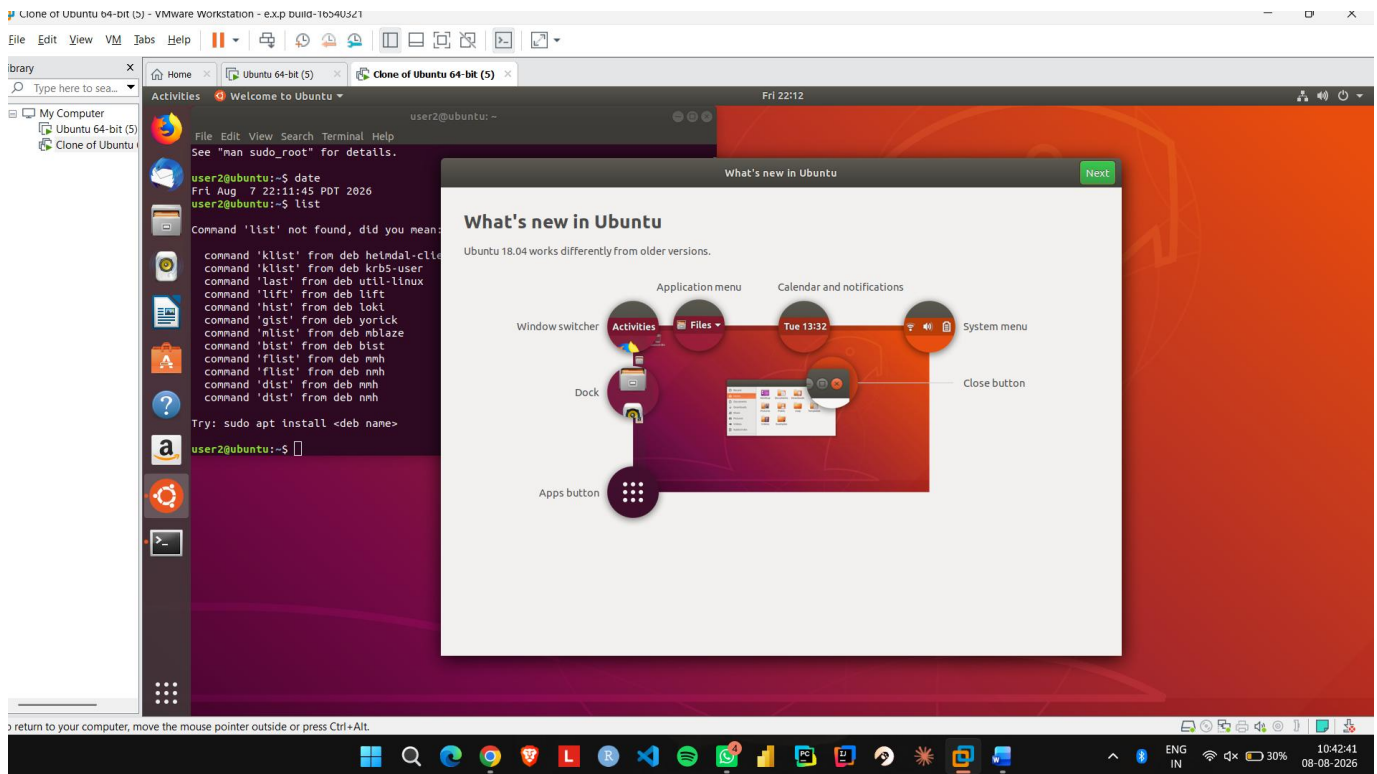
STEP 3: Select Full Clone and enter a name for the cloned VM.



STEP 4: Complete the cloning process and start the cloned VM.



STEP 5: Restore the previous snapshot, start the VM, and verify that the previous state is loaded.



Result:

The VM was successfully **cloned and tested**, and its previous saved state was successfully restored